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Exercise training device

Abstract

An exercise training device, including a main pole, a plurality of training arms disposed on at least a portion of the main pole to move in the first direction or the second direction in response to an application of force thereto, a plurality of collar clamps disposed between each of the plurality of training arms to secure each of the plurality of training arms to the main pole, and a plurality of ball bearings disposed between each of the plurality of training arms to facilitate movement of each of the plurality of training arms around an axis in a perpendicular direction with respect to the main pole.

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Background/Summary

BACKGROUND

1. Field

(1) The present general inventive concept relates generally to exercise, and particularly, to an exercise training device.

2. Description of the Related Art

(2) Martial arts are a system and ideology of combat used for different purposes, such as self-defense, competition, and/or exercise. Improvement in martial arts requires dedication and persistence in training.

(3) Many martial arts practitioners use a variety of means and/or tools to practice their skills. Moreover, practicing can include another practitioner and/or a training dummy. However, most training dummies are static structures with immovable parts while partners offer more dynamic training.

(4) Therefore, there is a need for an exercise training device that simulates training for martial arts with a live partner.

SUMMARY
(5) The present general inventive concept provides an exercise training device.
(6) Additional features and utilities of the present general inventive concept will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the general inventive concept.
(7) The foregoing and/or other features and utilities of the present general inventive concept may be achieved by providing an exercise training device, including a main pole, a plurality of training arms disposed on at least a portion of the main pole to move in the first direction or the second direction in response to an application of force thereto, a plurality of collar clamps disposed between each of the plurality of training arms to secure each of the plurality of training arms to the main pole, and a plurality of ball bearings disposed between each of the plurality of training arms to facilitate movement of each of the plurality of training arms around an axis in a perpendicular direction with respect to the main pole.
(8) The plurality of collar clamps may surround each of the plurality of training arms.
(9) The plurality of ball bearings may surround each of the plurality of training arms.
(10) The exercise training device may further include a plurality of connecting poles disposed between each of the plurality of training arms to connect at least one first of the plurality of collar clamps to at least one second of the plurality of collar clamps.
(11) The exercise training device may further include a base removably connected to the main pole to store at least one of sand and water therein to stabilize the main pole on an external surface.
(12) The base may include a container to receive at least one of the sand and water therein, and a cap removably connected to the container to prevent access within the container in a first position and allow access within the container in a second position.
(13) The exercise training device may further include a spring disposed on at least a portion of the container around the cap to move the main body back-and-forth in a springing motion in response to an application of force on at least one of the main body, the plurality of training arms, the plurality of collar clamps, and the plurality of ball bearings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) These and/or other features and utilities of the present generally inventive concept will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings of which:
(2) FIG. **1** illustrates a side perspective view of an exercise training device, according to an exemplary embodiment of the present general inventive concept.
DETAILED DESCRIPTION
(3) Various example embodiments (a.k.a., exemplary embodiments) will now be described more fully with reference to the accompanying drawings in which some example embodiments are illustrated. In the figures, the thicknesses of lines, layers and/or regions may be exaggerated for clarity.
(4) Accordingly, while example embodiments are capable of various modifications and alternative forms, embodiments thereof are shown by way of example in the figures and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but on the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure. Like numbers refer to like/similar elements throughout the detailed description.
(5) It is understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening

elements may be present. In contrast, when an element is referred to as being "directly connected" or "directly coupled" to another element, there are no intervening elements present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.).
(6) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises," "comprising," "includes" and/or "including," when used herein, specify the presence of stated features, integers, steps, operations, elements and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.
(7) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, e.g., those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art. However, should the present disclosure give a specific meaning to a term deviating from a meaning commonly understood by one of ordinary skill, this meaning is to be taken into account in the specific context this definition is given herein.
LIST OF COMPONENTS
(8) Exercise Training Device **100** Main Pole **110** Training Arms **120** First Arm **121** First Connecting Section **121***a* Second Arm **122** Second Connecting Section **122***a* Third Arm **123** Third Connecting Section **123***a* Collar Clamps **130** Ball Bearings **140** Connecting Poles **150** Base **160** Container **161** Cap **162** Spring **170** Knob **180**
(9) FIG. **1** illustrates a side perspective view of an exercise training device **100**, according to an exemplary embodiment of the present general inventive concept.
(10) The exercise training device **100** may be constructed from at least one of metal, plastic, wood, and rubber, etc., but is not limited thereto. Also, the exercise training device **100** may have foam thereon to absorb impact and prevent injury to a user. The exercise training device **100** may be specially constructed to emit sounds similar to fighting in response to receiving the impact thereon.
(11) The exercise training device **100** may include a main pole **110**, a plurality of training arms **120**, a plurality of collar clamps **130**, a plurality of ball bearings **140**, a plurality of connecting poles **150**, a base **160**, a spring **170**, and a knob **180**, but is not limited thereto.
(12) Referring to FIG. **1**, the main pole **110** is illustrated to have a cylindrical shape. However, the main pole **110** may be rectangular, circular, triangular, pentagonal, hexagonal, heptagonal, octagonal, or any other shape known to one of ordinary skill in the art, but is not limited thereto.
(13) The main pole **110** may be elongate and sturdy. Moreover, the main pole **110** may absorb an impact thereto, such as receiving a strike from a punch and/or a kick by the user.
(14) The plurality of training arms **120** may include a first arm **121**, a second arm **122**, and a third arm **123**, but is not limited thereto.
(15) The first arm **121** may be movably (i.e., rotatably) disposed on at least a portion of the main pole **110**. The first arm **121** may move (i.e., rotate) three hundred sixty (360) degrees in a first direction (i.e., clockwise) or a second direction (i.e., counterclockwise) in response to an application of force (e.g., punching, kicking, pushing, pulling) thereto. In other words, the first arm **121** may spin and/or rotate in response to being struck by the user during martial arts training.
(16) The first arm **121** may include a first connecting section **121***a*, but is not limited thereto.
(17) The first connecting section **121***a* may have a pole receiving aperture to receive a pole therethrough.
(18) The second arm **122** may be movably (i.e., rotatably) disposed on at least a portion of the main pole **110**. The second arm **122** may move (i.e., rotate) three hundred sixty (360) degrees in the first direction (i.e., clockwise) or the second direction (i.e., counterclockwise) in response to an

application of force (e.g., punching, kicking, pushing, pulling) thereto. In other words, the second arm **122** may spin and/or rotate in response to being struck by the user during martial arts training.
(19) The second arm **122** may include a second connecting section **122***a*, but is not limited thereto.
(20) The second connecting section **122***a* may have a pole receiving aperture to receive a pole therethrough.
(21) The third arm **123** may be movably (i.e., rotatably) disposed on at least a portion of the main pole **110**. The third arm **123** may move (i.e., rotate) three hundred sixty (360) degrees in the first direction (i.e., clockwise) or the second direction (i.e., counterclockwise) in response to an application of force (e.g., punching, kicking, pushing, pulling) thereto. In other words, the third arm **123** may spin and/or rotate in response to being struck by the user during martial arts training.
(22) The third arm **123** may include a third connecting section **123***a*, but is not limited thereto.
(23) The third connecting section **123***a* may have a pole receiving aperture to receive a pole therethrough.
(24) The first arm **121** may be disposed at a waist area of the user, the second arm **122** may be disposed at a shoulder area of the user, and/or the third arm **123** may be disposed at a head area of the user. The first arm **121**, the second arm **122**, and/or the third arm **123** may be distanced from each other.
(25) As such, the first arm **121**, the second arm **122**, and/or the third arm **123** may simulate practice of martial arts with a live opponent. More specifically, striking the first arm **121**, the second arm **122**, and/or the third arm **123** may cause each to move and attempt to strike the user during spinning. Thus, the user may practice martial arts skills, reflexes, and/or hand-eye coordination.
(26) The plurality of collar clamps **130** may be disposed between the main pole **110** and the first arm **121**, between the first arm **121** and the second arm **122**, between the second arm **122** and the third arm **123**, on an end surrounding the third arm **123**. Additionally, each of the plurality of collar clamps **130** may surround (i.e., on each side thereof) the first connecting section **121***a*, the second connecting section **122***a*, and/or the third connecting section **123***a*. Therefore, the first connecting section **121***a*, the second connecting section **122***a*, and/or the third connecting section **123***a* may receive at least one of the plurality of collar clamps **130** therethrough. The plurality of collar clamps **130** may connect and/or secure the first arm **121**, the second arm **122**, and/or the third arm **123** to the main pole **110**.
(27) The plurality of ball bearings **140** may be disposed between the main pole **110** and the first arm **121**, between the first arm **121** and the second arm **122**, between the second arm **122** and the third arm **123**, on the end surrounding the third arm **123**. Additionally, each of the plurality of ball bearings **140** may surround (i.e., on each side thereof) the first connecting section **121***a*, the second connecting section **122***a*, and/or the third connecting section **123***a*. Each of the plurality of ball bearings **140** may facilitate movement (i.e., rotation) of the first arm **121**, the second arm **122**, and/or the third arm **123** around an axis in a perpendicular direction with respect to the main pole **110**.
(28) The plurality of connecting poles **150** may be disposed between first arm **121** and the second arm **122**, and between the second arm **122** and the third arm **123**. Each of the plurality of connecting poles **150** may connect at least one first of the plurality of collar clamps **130** to at least one second of the plurality of collar clamps **130**. As such, the plurality of connecting poles **150** may extend a length and/or a height of the main pole **110**.
(29) The base **160** may include a container **161** and a cap **162**, but is not limited thereto.
(30) The container **161** may be removably disposed on an external surface (e.g., a ground surface, a table, a countertop, etc.) and/or removably connected to at least a portion of the main pole **110**. The container **161** may store sand and/or liquid (e.g., water) therein. As such, the container **161** may have a weight to prevent the main pole **110**, the plurality of arms **120**, the plurality of collar clamps **130**, the plurality of ball bearings **140**, and/or the plurality of connecting poles **160** from moving away and/or falling over. Accordingly, the container **161** may stabilize the main pole **110**, the

plurality of arms **120**, the plurality of collar clamps **130**, the plurality of ball bearings **140**, and/or the plurality of connecting poles **160**.

(31) The cap **162** may be removably connected (i.e., threadably screwed on) to the container **161**. The cap **162** may move (i.e., rotate) from closed on the container **161** in a first position to at least partially opened away from the container **161** in a second position. Conversely, the cap **162** may move (i.e., rotate) from opened away from the container **161** in the second position to closed on the container **161** in the first position. In other words, the cap **162** may prevent access within the container **161** in the first position and allow access within the container **161** in the second position. As such, the container **161** may receive the sand and/or the liquid while the cap **162** is removed.

(32) The spring **170** may be springingly disposed on at least a portion of the container **161** around the cap **162**. The spring **170** may be removed to allow access to the cap **162** as necessary. The spring **170** may allow the main pole **110**, the plurality of arms **120**, the plurality of collar clamps **130**, the plurality of ball bearings **140**, and/or the plurality of connecting poles **160** to move in response to the application of force thereon. More specifically, the main pole **110**, the plurality of arms **120**, the plurality of collar clamps **130**, the plurality of ball bearings **140**, and/or the plurality of connecting poles **160** may bounce back-and-forth in a springing motion in response to the application of force, such as being struck by the user during martial arts practice.

(33) The knob **180** may be movably (i.e., rotatably) disposed on at least a portion of the main pole **110**. The knob **180** may prevent removal of at least one of the plurality of collar clamps **130** within the main body **110** in response to moving (i.e., rotating) in a first direction (i.e., clockwise). Conversely, the knob **180** may allow removal of at least one of the plurality of collar clamps **130** from the main body **110** in response to moving (i.e., rotating) in a second direction (i.e., counterclockwise).

(34) Therefore, the exercise training device **100** may simulate training with a live partner by the user. Also, the exercise training device **100** may improve martial arts skills, cardio, mobility, and/or reflexes during practice through movement of the plurality of arms **120**.

(35) The present general inventive concept may include an exercise training device **100**, including a main pole **110**, a plurality of training arms **120** disposed on at least a portion of the main pole **110** to move in the first direction or the second direction in response to an application of force thereto, a plurality of collar clamps **130** disposed between each of the plurality of training arms **120** to secure each of the plurality of training arms **120** to the main pole **110**, and a plurality of ball bearings **140** disposed between each of the plurality of training arms **120** to facilitate movement of each of the plurality of training arms **120** around an axis in a perpendicular direction with respect to the main pole **110**.

(36) The plurality of collar clamps **130** may surround each of the plurality of training arms **120**.

(37) The plurality of ball bearings **140** may surround each of the plurality of training arms **120**.

(38) The exercise training device **100** may further include a plurality of connecting poles **150** disposed between each of the plurality of training arms **120** to connect at least one first of the plurality of collar clamps **130** to at least one second of the plurality of collar clamps **130**.

(39) The exercise training device **100** may further include a base **160** removably connected to the main pole **110** to store at least one of sand and water therein to stabilize the main pole **110** on an external surface.

(40) The base **160** may include a container **161** to receive at least one of the sand and water therein, and a cap **162** removably connected to the container **161** to prevent access within the container **161** in a first position and allow access within the container **161** in a second position.

(41) The exercise training device **100** may further include a spring **170** disposed on at least a portion of the container **161** around the cap **162** to move the main body **110** back-and-forth in a springing motion in response to an application of force on at least one of the main body **110**, the plurality of training arms **120**, the plurality of collar clamps **130**, and the plurality of ball bearings **140**.

(42) Although a few embodiments of the present general inventive concept have been shown and described, it will be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the general inventive concept, the scope of which is defined in the appended claims and their equivalents.

Claims

1. An exercise training device, comprising: a main pole; a plurality of training arms disposed on at least a portion of the main pole to move in the first direction or the second direction in response to an application of force thereto; a plurality of collar clamps disposed between each of the plurality of training arms to secure each of the plurality of training arms to the main pole; a plurality of ball bearings disposed between each of the plurality of training arms to facilitate movement of each of the plurality of training arms around an axis in a perpendicular direction with respect to the main pole; and a base disposed at a bottom-most portion of the exercise training device and configured to be removably connected to the main pole to store at least one of sand and water therein to stabilize the main pole on an external surface, the base comprising: a container to receive at least one of the sand and water therein, a cap removably connected to the container to prevent access within the container in a first position and allow access within the container in a second position, and a spring to directly contact a top portion of the container and to be disposed around the cap to move the main body back-and-forth in a springing motion in response to an application of force on at least one of the main body, the plurality of training arms, the plurality of collar clamps, and the plurality of ball bearings.
2. The exercise training device of claim 1, wherein the plurality of collar clamps surrounds each of the plurality of training arms.
3. The exercise training device of claim 1, wherein the plurality of ball bearings surrounds each of the plurality of training arms.
4. The exercise training device of claim 1, further comprising: a plurality of connecting poles disposed between each of the plurality of training arms to connect at least one first of the plurality of collar clamps to at least one second of the plurality of collar clamps.